Paul Mancion

Flexible CA/USA | Portfolio: paulmancion.com | linkedin.com/in/paulmancion
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Education

University of Guelph — Sep 2021 – May 2026
Honours. Bachelor of Computing, Computer Science (GPA 3.95/4.0) — *Guelph, ON*

University of Waterloo and Wilfrid Laurier — Sep 2020 – Apr 2021
Bachelor of Computer Science and Bachelor of Business Administration — *Waterloo, ON*

Technical Skills

Languages: Python, C, Java, SQL, JavaScript, TypeScript, Dart, HTML/CSS, Racket
Technologies: React, Next.js, Node.js, Flask, REST APIs, AWS, GCP, PostgreSQL, Docker, Vercel, Git
Bilingual: English, French

Projects

Autopilot | AI-powered quant trading platform | *Python, TypeScript, scikit-learn, Hugging Face, CCXT, AWS*

- Built and validated an **end-to-end ML** trading pipeline that matched/outperformed the original strategy through signal prediction, **hyperparameter tuning**, and **backtesting**.
- Building a **full-stack** trading automation platform where user-prompts drive the strategy creation, optimization, and deployment, using AST validation, **agent-facing MCP servers**, a React **frontend** and **Vercel** + **AWS** serverless stack.

Parkd App | P2P parking marketplace | *Flutter, Dart, Supabase/PostgreSQL, Mapbox, Stripe*

- Delivered an **MVP** with real-time booking, Mapbox geolocation, Stripe payments, and a Flutter **frontend** connected to Supabase/PostgreSQL **backend** services.
- Followed an **Agile development process** in a 4-person team using feature ownership, code review, shared components, and biweekly check-ins to maintain UI/UX consistency across core workflows.

Games Platform | Multiplayer platform & engines | *C, Python, TypeScript, SWIG, Supabase/PostgreSQL*

- Built a full-stack multiplayer web platform for 2 browser games with online PvP, matchmaking, chat, sessions and replays, integrating a TypeScript frontend with **Supabase** backend services and C game engines via Python/SWIG.
- Rewrote chess-variant engine in C, increasing move generation speed by **~30x**, using **alpha-beta Negamax**, bitboards, Zobrist hashing, **magic attacks**, transposition tables, and **custom heuristics**, and built a C 8-ball physics engine.

Orangepill | Nostr relay services | *Node.js, Nostr/Nostream, Bitcoin Lightning, LNbits, SQLite, Docker*

- Scaled a Nostr relay service to **2,000+** users and **millions** of relay events through multiple locations, authentication, Bitcoin **payments**, pubkey rate-limits, and **Dockerized** infrastructure.

Work Experience

Marcatus QED — Mar 2023 – May 2023
Marketing Communications Intern — *Toronto, ON*

- Created **Marcatus Mobile Education Platform** deck, showing impact across **77,000+** smallholder farmers.
- Developed the Marcatus **Unilever Global Summit 2023** deck, presenting expansion plans for **6,000+** farmers.

Docushield — May 2022 – Nov 2022
Product Growth Lead and Software Engineer Intern — *Los Angeles, CA*

- Drove **product-market fit** research, community initiatives, and IDO campaigns supporting a **$1M+** funding.
- Delivered **MVP** and **beta** releases for the core app and NFT platform by leading product design, **collaborating** with engineers across cross-functional teams, coordinating community **testing**, and supporting **CI/CD** release workflows.

Highschool Math & Science Tutor — Jun 2020 – Sep 2024

- Improved student understanding through personalized lessons clearly explaining concepts with practice exercises.